Ball rolling: what EIT adds

Simulation follow-up • 22 October 2026

The study contains 329 evaluation attempts with frozen schedules and sensor seeds excluded from 68 development attempts. It tests progress feedback, contact migration, reference dynamics, and sensing quality. All results are synthetic MuJoCo/FEM simulations; the physical EIT palm has not been validated by this study.

Main findings

Question	Result	Interpretation
Does ongoing EIT position feedback help?	After a +4 N, 0.35 s push, late error is 2.73 mm with palm odometry and 0.41 mm with EIT. Each condition has 8 trials.	EIT corrects progress after slip. Both conditions retain contact sensing and centre attachment.
Does updating the attachment help?	At 0.3 Nm/rad, mean peak pitch is 4.01° for a latched EIT attachment and 1.92° for an updated one.	Contact migration makes continuous updates relevant to palm alignment. Load and completion must be considered separately.
Is the carriage generally better?	Stop–restart overshoot falls from 0.99 to 0.14 mm with the carriage, while integrated error rises from 10.12 to 13.03 mm s.	The tested carriage trades responsiveness for smaller overshoot; this is not a general superiority claim.

What this can strengthen in the paper

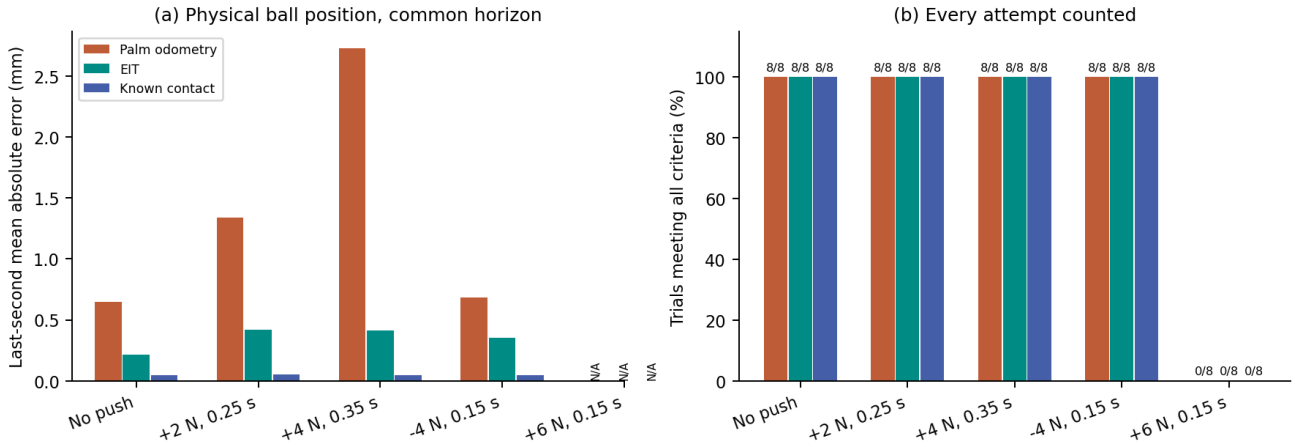
The strongest candidate is EIT correction of virtual task state after an unobserved disturbance, together with attachment updates that limit unwanted palm rotation. This extends the evidence beyond initial contact detection. A benefit also obtained with the prescribed-reference controller supports the sensing contribution rather than proving a unique advantage of carriage dynamics.

The full electrical pipeline runs online: synthetic 40-channel FEM voltages feed the restored 100-tree Extra Trees position estimator. Its force output is excluded from feedback. No new model is trained for these experiments.

1. Ongoing progress feedback after a disturbance

Why. Palm travel predicts ball travel only while the assumed rolling relation remains accurate. External forces and slip can break that relation. This experiment tests whether tactile position measurements correct the resulting task-state error.

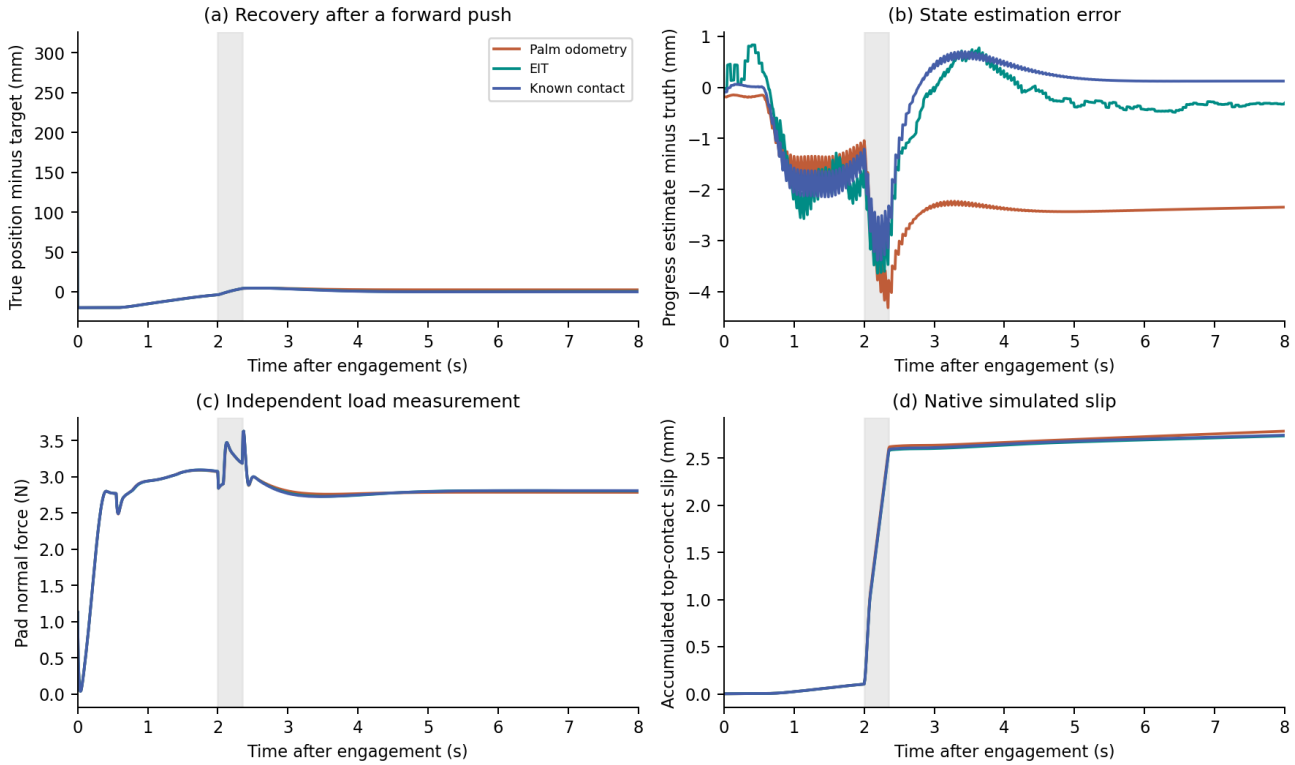
What and how. Request a 20 mm roll; apply a horizontal ball-body force two seconds after engagement without notifying the controller. Compare palm odometry, decoded EIT contact position transformed into world coordinates, and true contact location at the same sample times. All have centre attachment, contact events, carriage gains, limits, and camera initialization in common. Two landing offsets (0 and +10 mm) and four sensor seeds give eight trials per condition.



Scoring. Run for eight seconds after initial engagement. Score the mean absolute true ball error and speed in the final second. Require error ≤5 mm, speed ≤4 mm/s, cross-path error ≤5 mm, peak pad force ≤12 N, rim force <0.2 N, and no protective stop or premature engagement. Guarded attempts remain failures; missing late windows are not imputed as zero.

Forward disturbance. With +4 N for 0.35 s, completion is 8/8 for odometry and 8/8 for EIT. Mean peak normal loads are 3.61 and 3.61 N. The accuracy improvement is 84.9%. Nominal, reverse, and severe conditions remain visible to bound this finding.

2. Why the feedback result occurs

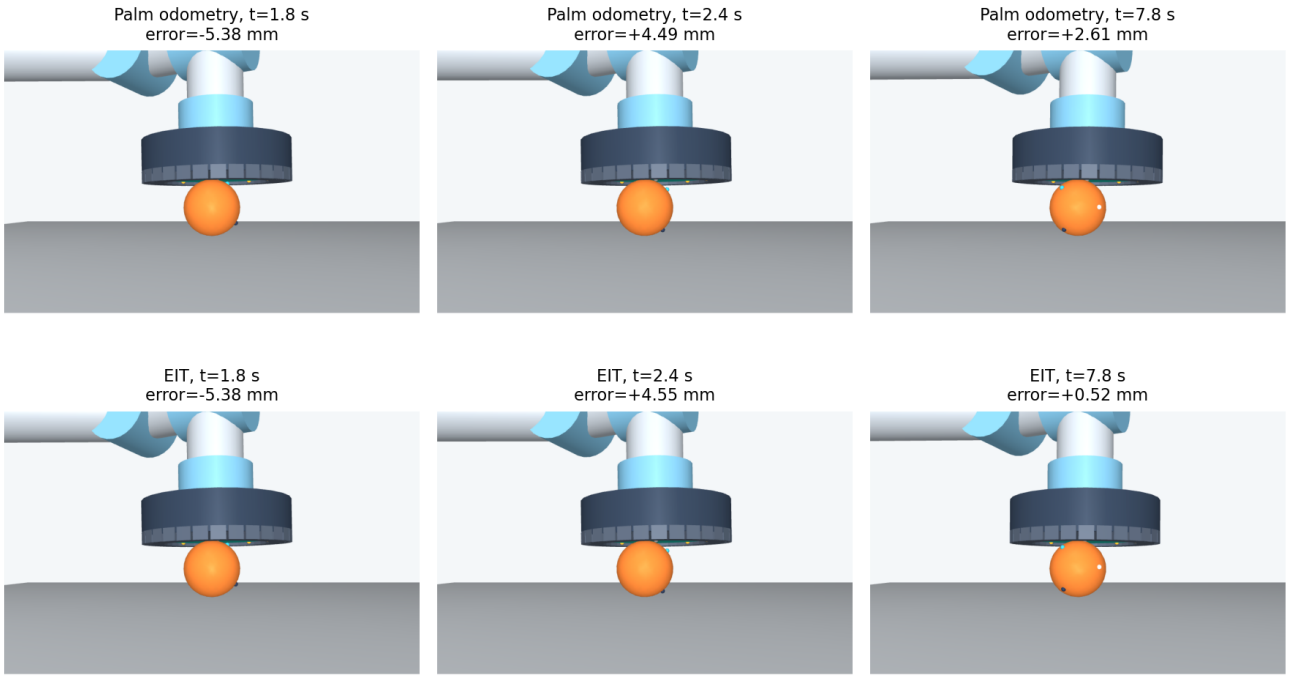


The example is the first evaluation seed (1600), +10 mm landing, and +4 N push. The grey interval is the imposed force. Actual slip is measured from native contact kinematics; it is not commanded as a displacement. A persistent odometry error can remain after the ball stops. EIT updates the progress estimate and the virtual coordinate, with reference rebasing to preserve the instantaneous coupling extension.

Across forward-push trials, accumulated upper-contact slip averages 2.66 mm with odometry and 2.63 mm with EIT. EIT corrects the consequence of slip; this does not establish slip prevention. The controller receives no physical ball position, rotation, external-force value, or disturbance timestamp.

The prescribed-reference check gives 2.23 mm with odometry and 0.46 mm with EIT, with completions 8/8 and 8/8. This identifies whether the sensing benefit extends beyond the virtual carriage.

3. Simulated physical setup and motion



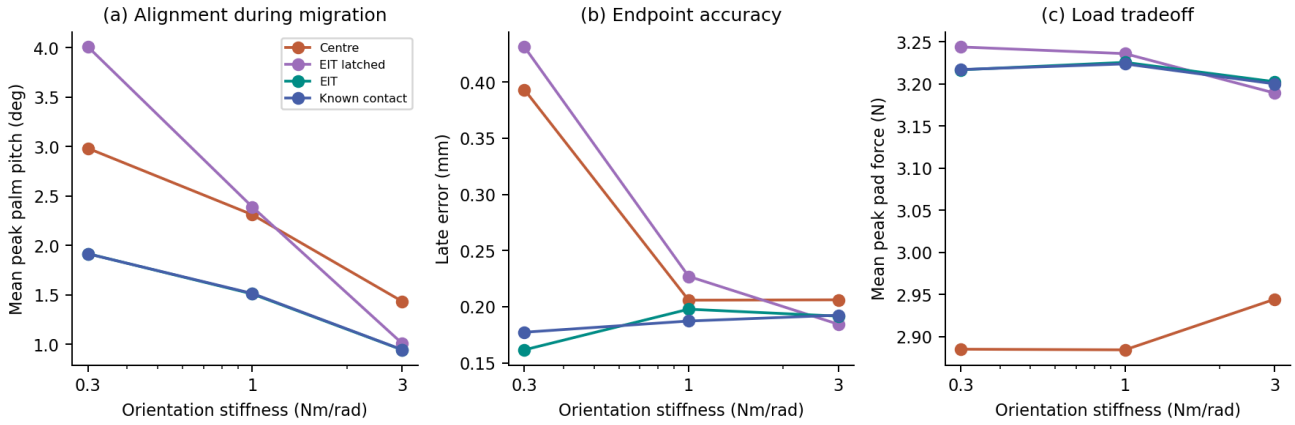
These are actual saved MuJoCo states for the same two feedback conditions, before the push, shortly after it, and near the final scoring window. The palm is above a free sphere on a horizontal table. The images show the modeled geometry; endpoint differences are quantified by the independent position traces rather than inferred from image appearance.

The assumed active pad diameter is 90 mm. The housing opening radius is 47 mm with a 3 mm recess. The sphere diameter is 50 mm and mass approximately 72 g. Nominal friction is 0.7 and preload demand is 3 N. The arm runs with ideal torque actuation and exact bias compensation at 1 ms. No regrasp occurs in the short-stroke comparisons; geometric limits are reported separately.

4. Attachment updates during contact migration

Why. A contact point latched at engagement can become stale as the ball rolls across the palm. A virtual force acting at that stale point produces a different moment. This experiment tests palm alignment and task error as orientation compliance changes.

What and how. Hold EIT progress feedback and event handling fixed. Compare centre attachment, EIT attachment latched once, EIT attachment updated during rolling, and updated true contact. Cross +10 and +25 mm offsets, orientation stiffnesses 0.3/1/3 Nm/rad, and four new sensor seeds. Every comparison matches stiffness, damping, and limits.



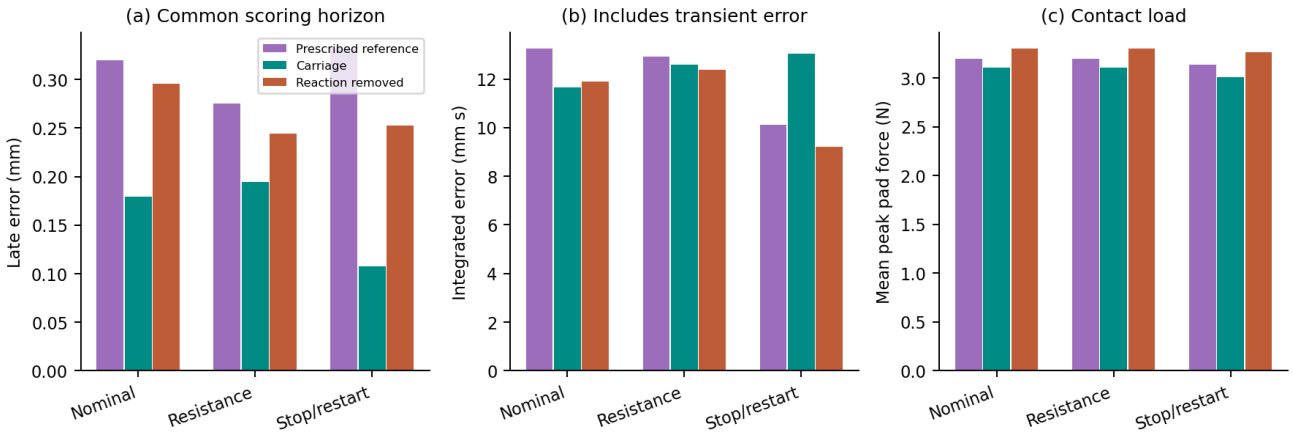
At 0.3 Nm/rad, mean peak pitch is 2.98° for centre, 4.01° for latched EIT, and 1.92° for updated EIT. The update reduces pitch by 52.2% relative to latching. Corresponding mean peak loads are 2.89, 3.24, and 3.22 N. These are alignment and load measurements; no new pitch success threshold was selected after seeing results.

A separate -25 mm initial offset exposes the finite pad boundary during a +20 mm continuous roll. Its failures are retained and cannot be repaired merely by estimating the contact more accurately. Regrasp would require a different task protocol.

5. Does carriage reaction improve performance?

Why. The previous endpoint comparison stopped partly according to virtual-state velocity, which differed between controllers. Common physical scoring now tests whether that earlier advantage persists.

What and how. Compare the prescribed reference, carriage, and the same carriage with its coupling-reaction term removed. Test nominal rolling, temporary spring resistance (2.5–3.3 s after engagement), and a 10 mm target followed by a 20 mm target at 3.5 s. Four new seeds per cell; common EIT progress, centre attachment, coupling gains, and force/speed limits. Internal parameters are inherited, with no claim of optimal tuning.



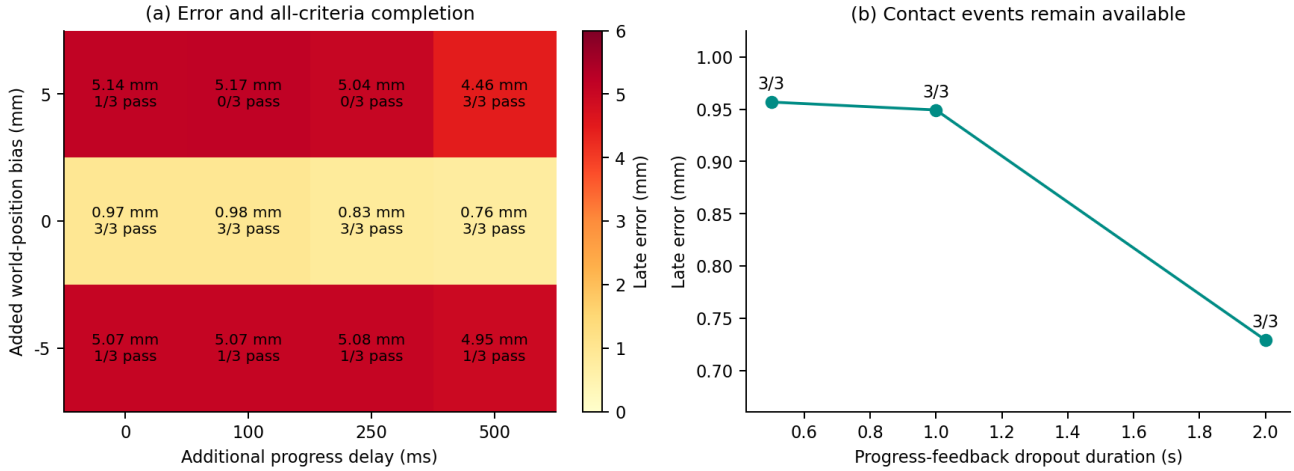
Condition	Prescribed error	Carriage error	Reaction removed
nominal	0.32 mm	0.18 mm	0.30 mm
resistance	0.27 mm	0.19 mm	0.24 mm
stop_restart	0.33 mm	0.11 mm	0.25 mm

In the stop–restart evaluation, carriage mean overshoot is 0.14 mm versus 0.99 mm for the prescribed reference and 2.62 mm with reaction removed. Carriage late error is 0.11 versus 0.33 mm, but integrated error increases from 10.12 to 13.03 mm s and settling occurs later (4.85 versus 4.66 s after engagement). This supports an overshoot–responsiveness tradeoff for these inherited gains. The single-seed pilot did not consistently favour the carriage, so the small evaluation cannot establish general superiority. Removing reaction isolates one mechanism component; it is not a tuned alternative controller.

6. Sensing quality versus task performance

Why. A useful training or acquisition requirement must depend on closed-loop consequences. This study isolates errors in the ball-progress stream while retaining the same EIT contact detector and centre attachment.

What and how. Under the +4 N, 0.35 s push at +10 mm landing, add world-position biases of -5/0/+5 mm and transport delays of 0/100/250/500 ms. Use three new sensor seeds per cell. Convert contact position with the pose at sampling before delay. Separately suppress progress updates for 0.5/1/2 s beginning at the push; contact events continue.



With no added bias, all three trials pass at each tested additional delay through 500 ms. At zero added delay, either sign of 5 mm bias leaves only 1/3 passing, with mean late error about 5.1 mm. All progress-dropout conditions pass 3/3 after updates resume. Delay effects are nonmonotonic in this slow, finite-duration task; the results do not show that latency is beneficial or supply a maximum safe delay. Bias near the task tolerance is the clearer limitation. Contact detection remained available, so progress dropout does not test physical loss recovery.

Training should cover curved and migrating contacts if their measured errors exceed the useful region. Timestamp alignment and acquisition speed should be prioritized when delay is limiting. Real curved-object calibration is still required before overlaying a physical sensor on this map. No physical sensing operating point is claimed here.

7. Provenance, checks, and claim limits

The evaluation contains 329 attempted trials and 2,709,669 physics steps. Saved configurations match the frozen schedule, the recorded source hashes match, and the trajectory audit reports 0 issues. Baseline equivalence with all new options disabled gives zero difference in original trace columns and robot/object positions.

The initial 48 pilots used seed 1500, and 20 force-screen pilots used seed 1501. Evaluation selection followed these pilots. Evaluation seeds are 1600–1603 for progress, 1700–1703 for attachments, 1800–1803 for dynamics, and 1900–1902 for quality. They vary synthetic electrical sessions; counts are descriptive rather than independent manufactured-sensor evidence.

The table-translation pilot produced identical trajectories to nominal because moving the mocap position did not impose the intended horizontal contact-surface velocity. It is retained as an ineffective perturbation and excluded from disturbance claims. Gentle low-friction pilots also showed no material benefit. Severe-force and edge-contact failures are retained in the evaluation.

The synthetic electrical footprint is not calibrated for sphere curvature, indentation, shear, or physical material drift. Force prediction remains outside feedback. Native rigid-body contact approximates a soft interface. Full arm clearance, hardware torque-interface feasibility, and physical bracket dimensions remain unvalidated.

Halving the physics step to 0.5 ms in four representative feedback and attachment trials preserves all four outcomes. Maximum changes are 0.00493 mm in late error, 0.00243 degrees in peak pitch, and 0.00926 N in peak force. These checks support numerical consistency within the assumed model.

Recommended physical demonstration

Begin with an occluded short roll and a measured, repeatable forward push. Compare ongoing EIT progress against palm-motion odometry while sharing contact detection, initial camera information, attachment, and controller gains. Record independent ball motion, rotation, and contact load. Then compare latched and continuously updated attachments at matched orientation compliance. Retain the prescribed-reference control to show which benefits arise from sensing and which from mechanism dynamics.

Files

evaluation/protocol.json fixes the schedule and sources; evaluation/trials contains every saved state and trace; analyzed_metrics.csv and summary.json contain all-attempt results; audit.json records integrity checks. EVALUATION_DESIGN.md documents selection and scoring. The original 540-trial campaign is a separate evidence set.